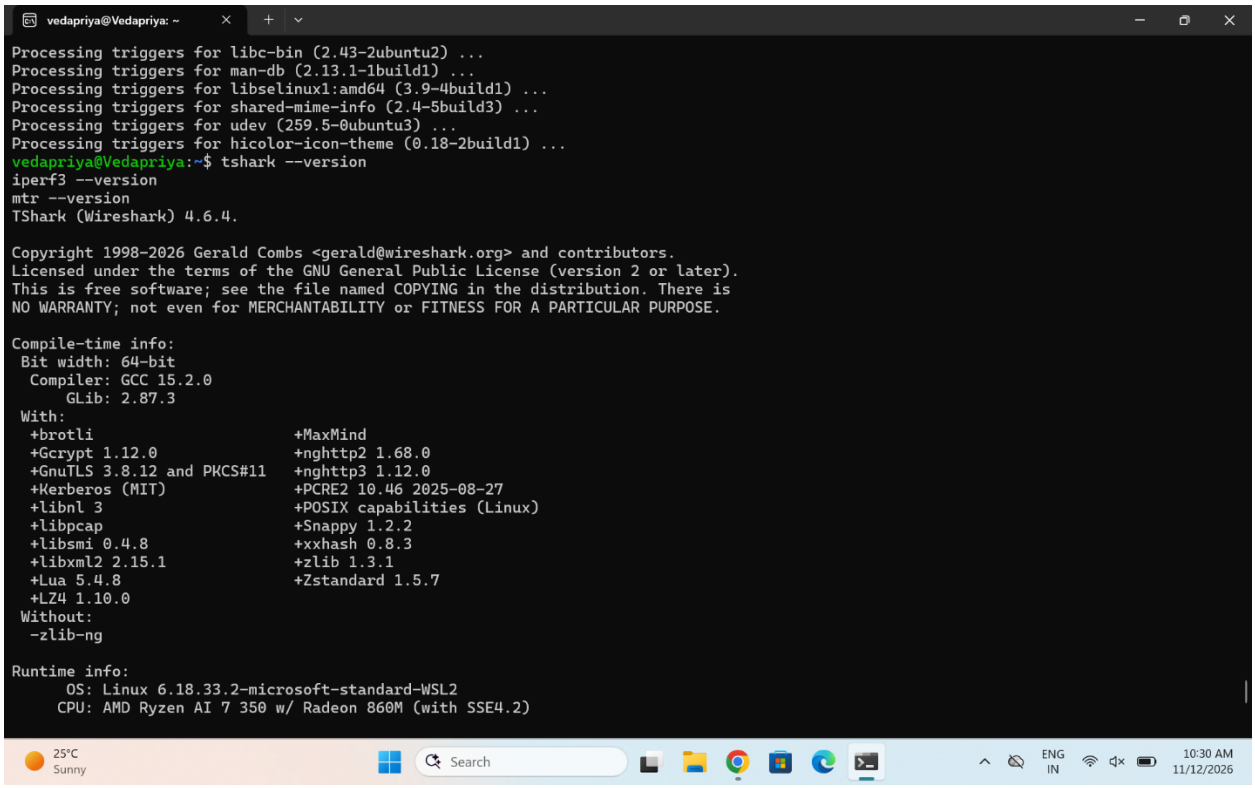
CNNP- Skill 2

THREE-POINT CAPTURE – Analyze packet captures from three points in a network path to understand packet flow and identify performance differences.

STAGE 1 — Install tools

Open Ubuntu and type the following commands in the Ubuntu terminal

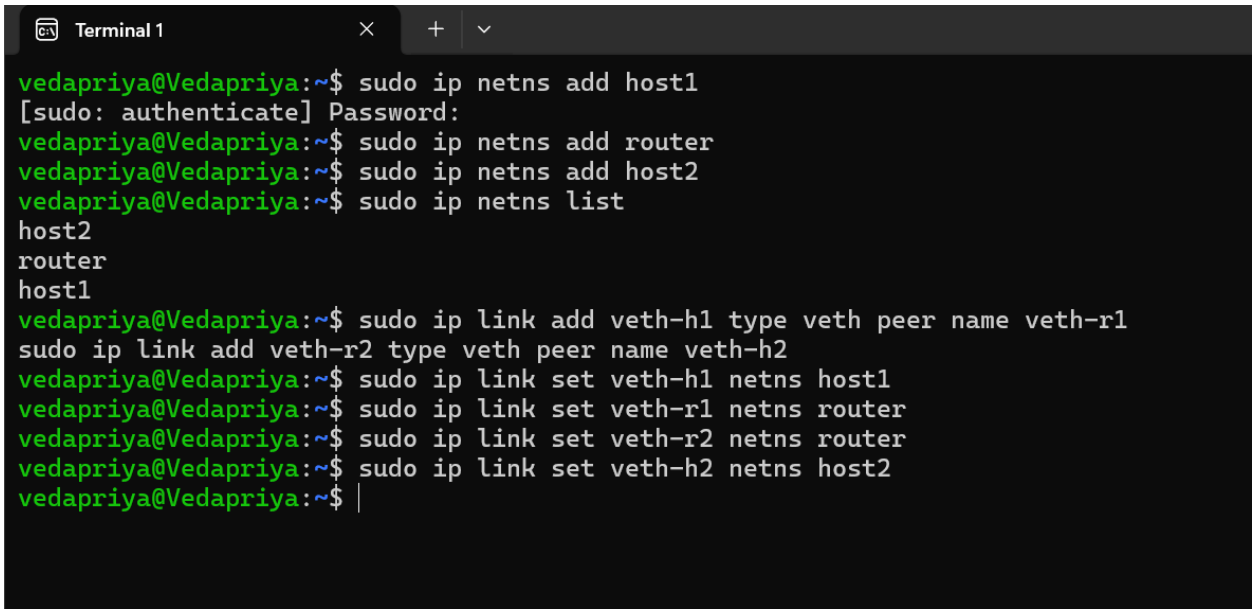
sudo apt update
sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd



STAGE 2 — Build the network

Open again ubuntu and rename that tab as Terminal 1 and run all of these one by one:

sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2



sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2

```
vedapriya@Vedapriya:~$ sudo ip netns add host1
[sudo: authenticate] Password:
vedapriya@Vedapriya:~$ sudo ip netns add router
vedapriya@Vedapriya:~$ sudo ip netns add host2
vedapriya@Vedapriya:~$ sudo ip netns list
host2
router
host1
vedapriya@Vedapriya:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
vedapriya@Vedapriya:~$ sudo ip link set veth-h1 netns host1
vedapriya@Vedapriya:~$ sudo ip link set veth-r1 netns router
vedapriya@Vedapriya:~$ sudo ip link set veth-r2 netns router
vedapriya@Vedapriya:~$ sudo ip link set veth-h2 netns host2
vedapriya@Vedapriya:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
vedapriya@Vedapriya:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
vedapriya@Vedapriya:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
vedapriya@Vedapriya:~$
```

sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router

```
sudo ip link set veth-r2 netns router
```

sudo ip link set veth-h2 netns host2

```
vedapriya@Vedapriya:~$ sudo ip netns add host1
[sudo: authenticate] Password:
vedapriya@Vedapriya:~$ sudo ip netns add router
vedapriya@Vedapriya:~$ sudo ip netns add host2
vedapriya@Vedapriya:~$ sudo ip netns list
host2
router
host1
vedapriya@Vedapriya:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
vedapriya@Vedapriya:~$ sudo ip link set veth-h1 netns host1
vedapriya@Vedapriya:~$ sudo ip link set veth-r1 netns router
vedapriya@Vedapriya:~$ sudo ip link set veth-r2 netns router
vedapriya@Vedapriya:~$ sudo ip link set veth-h2 netns host2
vedapriya@Vedapriya:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
vedapriya@Vedapriya:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
vedapriya@Vedapriya:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
vedapriya@Vedapriya:~$
```

STAGE 3 — Configure IP addresses (Note: In terminal 1 Only)

host1:

sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1

sudo ip netns exec host1 ip link set veth-h1 up

sudo ip netns exec host1 ip link set lo up

```
vedapriya@Vedapriya:~$ sudo ip link set veth-h1 netns host1
vedapriya@Vedapriya:~$ sudo ip link set veth-r1 netns router
vedapriya@Vedapriya:~$ sudo ip link set veth-r2 netns router
vedapriya@Vedapriya:~$ sudo ip link set veth-h2 netns host2
vedapriya@Vedapriya:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
vedapriya@Vedapriya:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
vedapriya@Vedapriya:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
vedapriya@Vedapriya:~$
```

router:

sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up

```
vedapriya@Vedapriya:~$ sudo ip link set veth-h1 netns host1
vedapriya@Vedapriya:~$ sudo ip link set veth-r1 netns router
vedapriya@Vedapriya:~$ sudo ip link set veth-r2 netns router
vedapriya@Vedapriya:~$ sudo ip link set veth-h2 netns host2
vedapriya@Vedapriya:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
vedapriya@Vedapriya:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
vedapriya@Vedapriya:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
vedapriya@Vedapriya:~$
```

host2:

sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up

```
vedapriya@Vedapriya:~$ sudo ip link set veth-h1 netns host1
vedapriya@Vedapriya:~$ sudo ip link set veth-r1 netns router
vedapriya@Vedapriya:~$ sudo ip link set veth-r2 netns router
vedapriya@Vedapriya:~$ sudo ip link set veth-h2 netns host2
vedapriya@Vedapriya:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
vedapriya@Vedapriya:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
vedapriya@Vedapriya:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
vedapriya@Vedapriya:~$
```

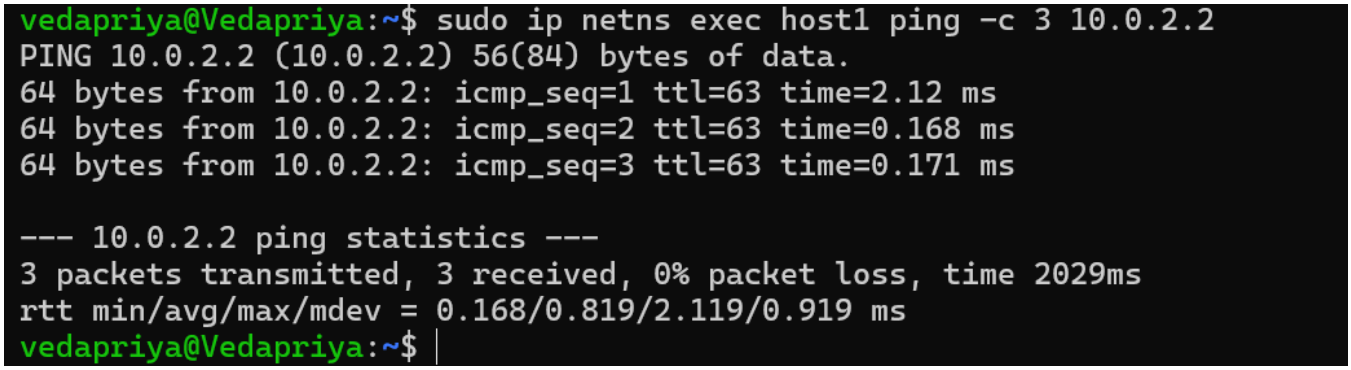
STAGE 4 — Enable routing

sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1

```
vedapriya@Vedapriya:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
vedapriya@Vedapriya:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
vedapriya@Vedapriya:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
vedapriya@Vedapriya:~$
```

STAGE 5 — Test connectivity

sudo ip netns exec host1 ping -c 3 10.0.2.2



You must see:

3 packets transmitted, 3 received, 0% packet loss

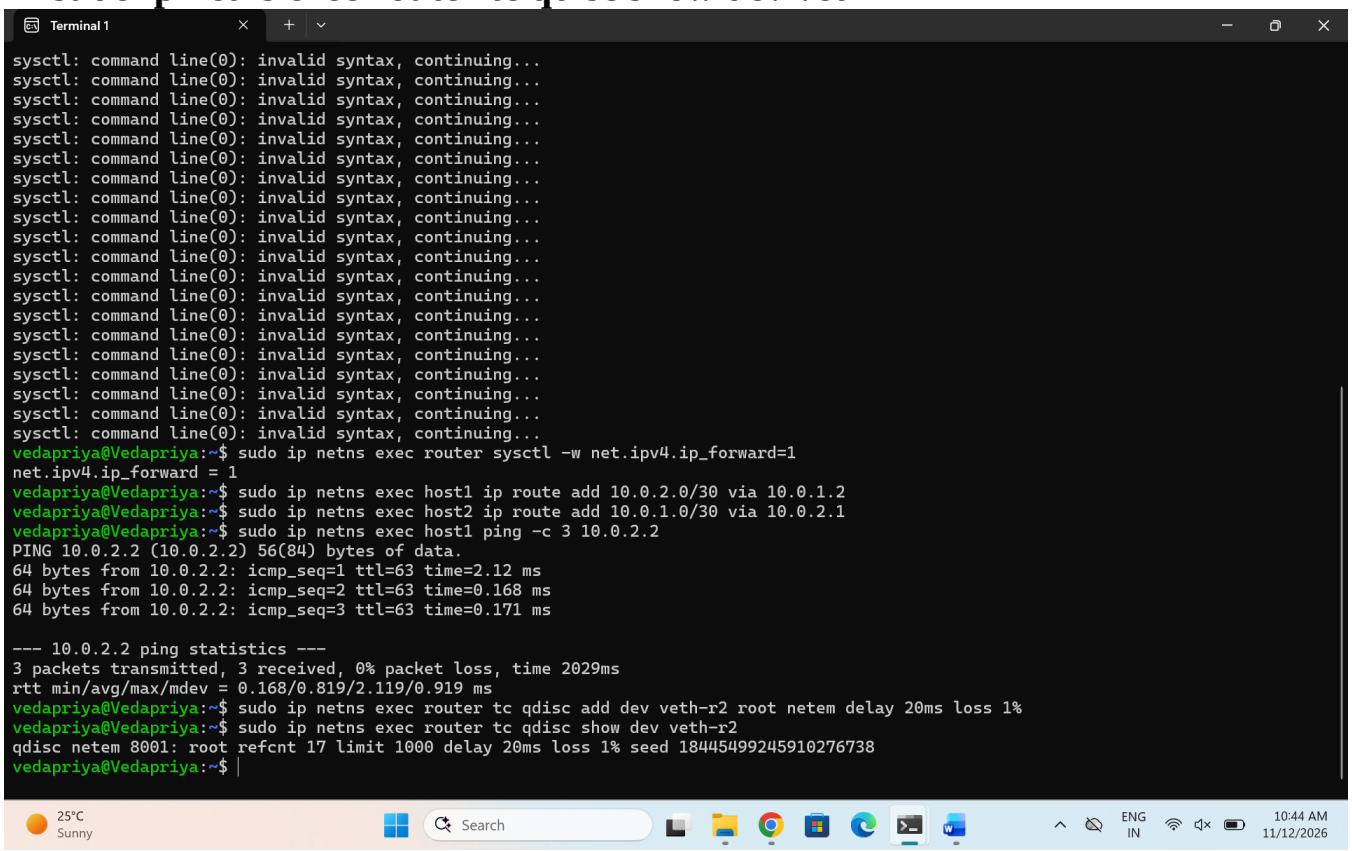
STAGE 6 — Add delay and loss

```
sysctl: command line(0): invalid syntax, continuing...
vedapriya@Vedapriya:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
vedapriya@Vedapriya:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
vedapriya@Vedapriya:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
vedapriya@Vedapriya:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=2.12 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.168 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.171 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2029ms
rtt min/avg/max/mdev = 0.168/0.819/2.119/0.919 ms
vedapriya@Vedapriya:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
vedapriya@Vedapriya:~$ sudo ip netns exec router tc qdisc show dev veth-r2
```

Verify:

sudo ip netns exec router tc qdisc show dev veth-r2

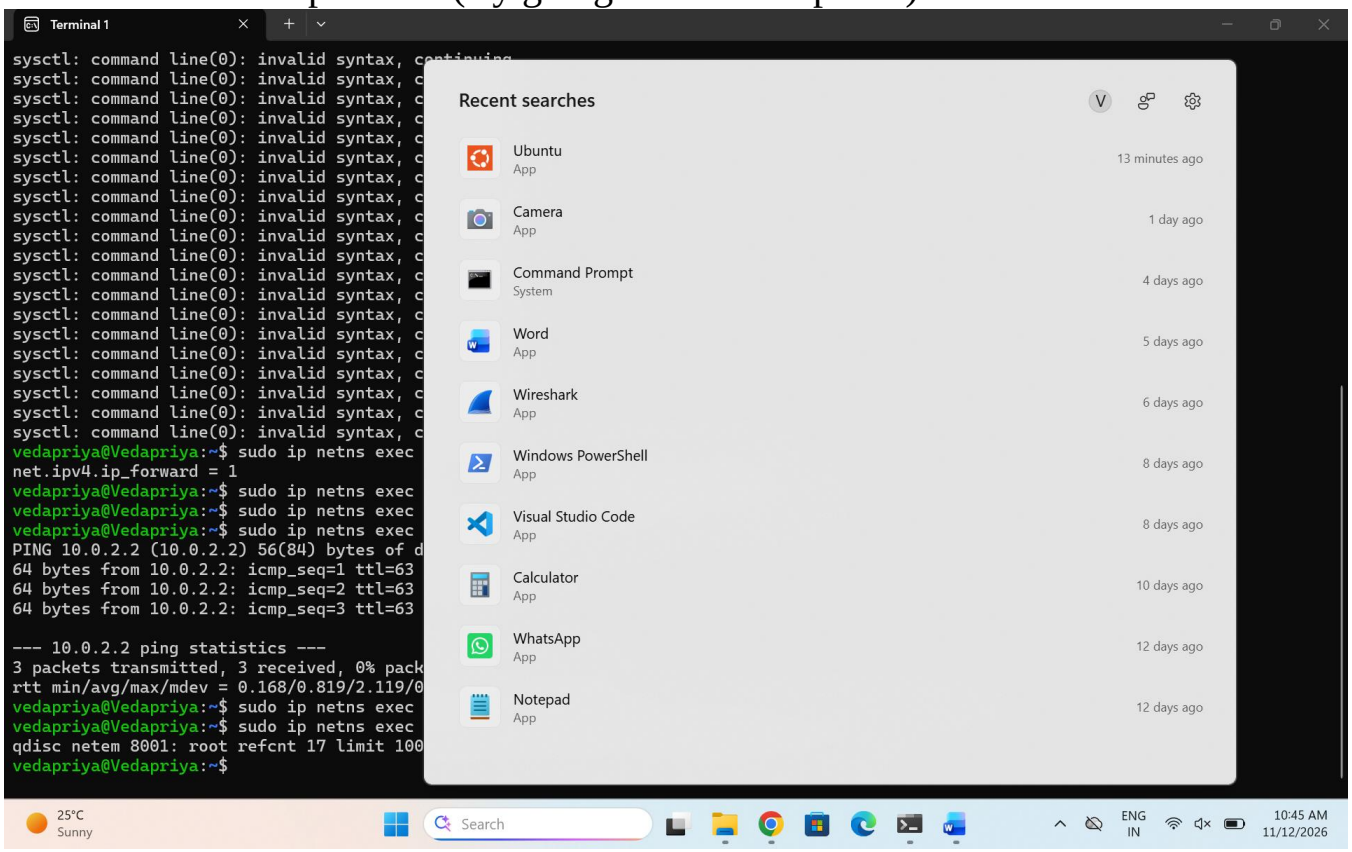


You must see netem delay 20ms loss 1%

STAGE 7 — Open 4 terminals

You need 4 terminals open now.

Terminal 1 — already open (used above)
Terminal 2 — open new (By going to Ubuntu option)



Terminal 3 — open new (By going to Ubuntu option)
Terminal 4 — open new (By going to Ubuntu option)

STAGE 8 — Start three captures simultaneously

Terminal 1 — capture at host1

sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap

```
vedapriya@Vedapriya:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
```

Terminal 2 — capture at router

sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap

```
vedapriya@Vedapriya:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
```

Terminal 3 — capture at host2

sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap

```
vedapriya@Vedapriya:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
```

All three show:

Capturing on 'veth-h1'

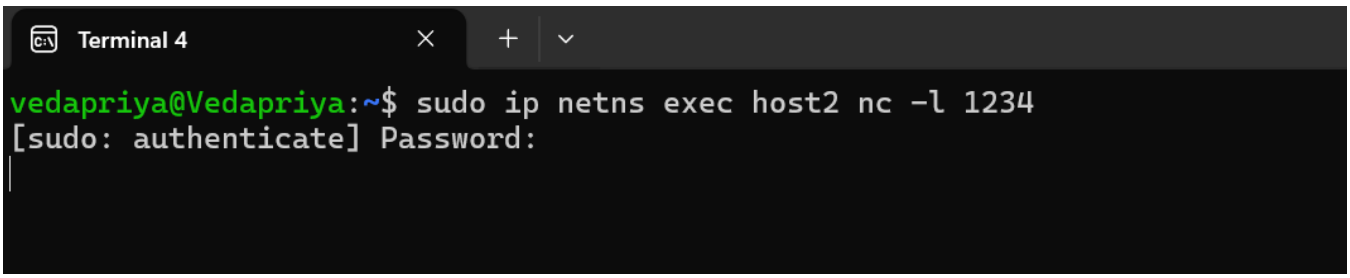
Leave all three running.

STAGE 9 — Send real message from host1 to host2

Go to Terminal 4.

First start the receiver on host2:

sudo ip netns exec host2 nc -l 1234



Open a **5th terminal.**

Connect from host1:

sudo ip netns exec host1 nc 10.0.2.2 1234

Now type this message and press Enter:

Hello from host1!

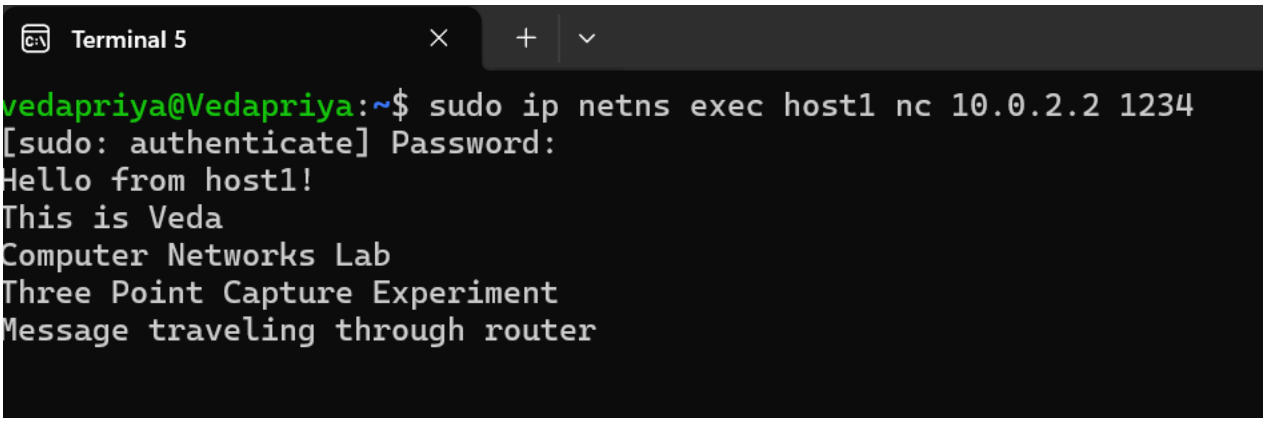
Type more messages:

This is Veda

Computer Networks Lab

Three Point Capture Experiment

Message traveling through router



STAGE 10 — What you see in real time

Terminal 4 (host2) shows your message arriving:

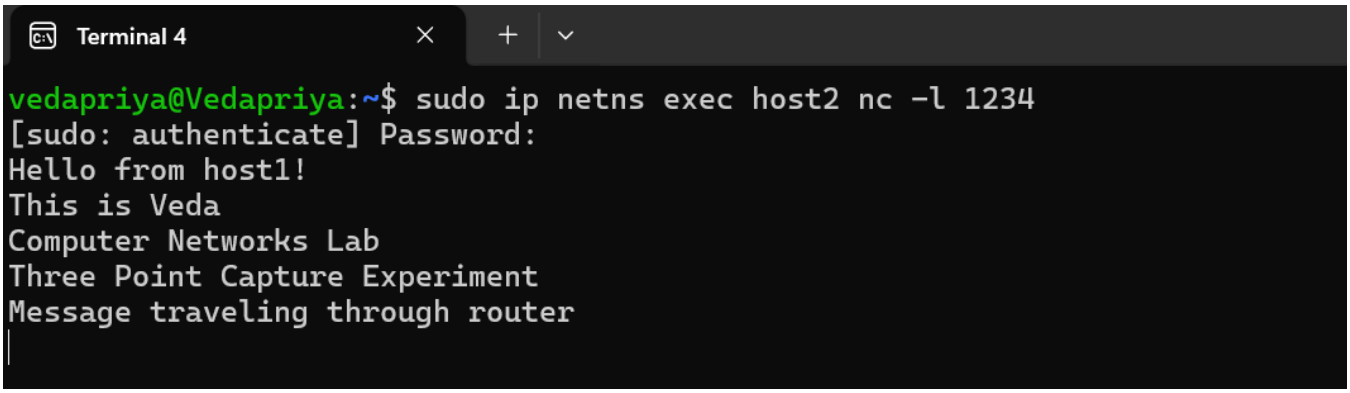
Hello from host1!

This is Veda

Computer Networks Lab

Three Point Capture Experiment

Message travelling through router



Terminal 1, 2, 3 (tshark) show packet counts increasing:

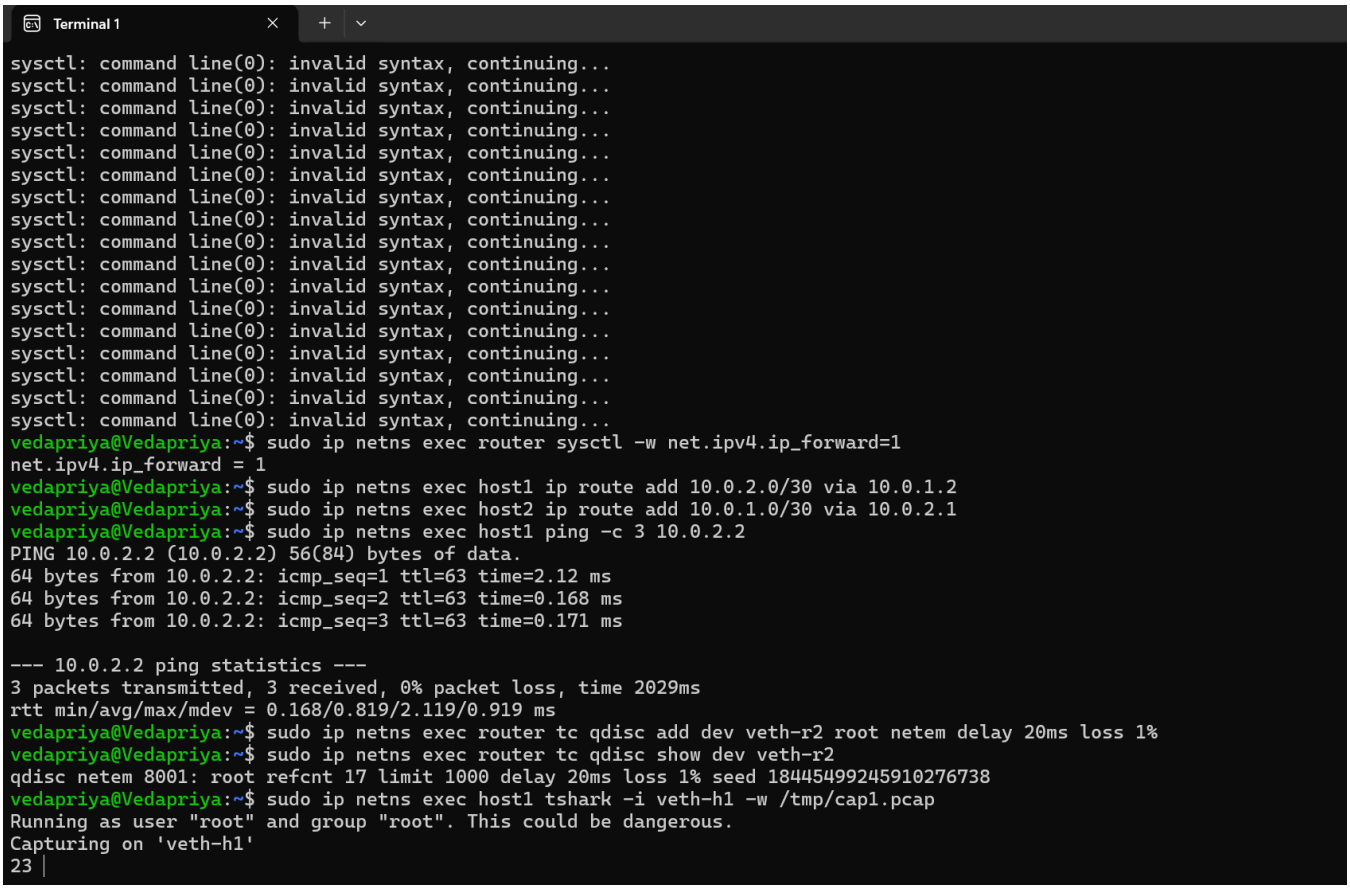
Capturing on 'veth-h1'

10 packets

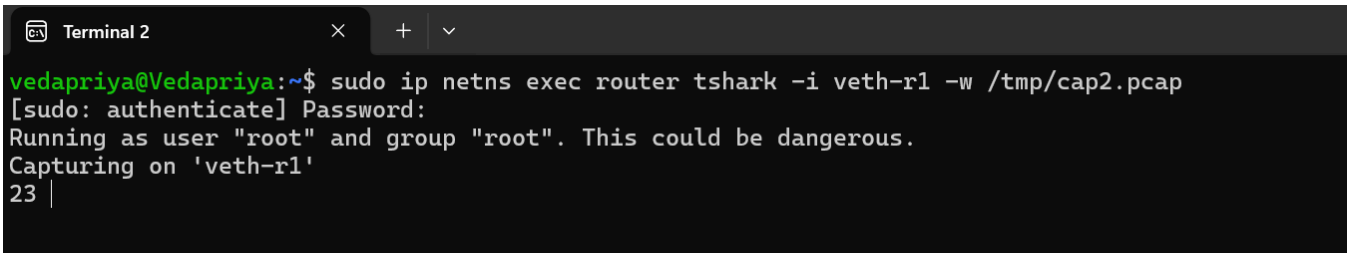
24 packets

31 packets

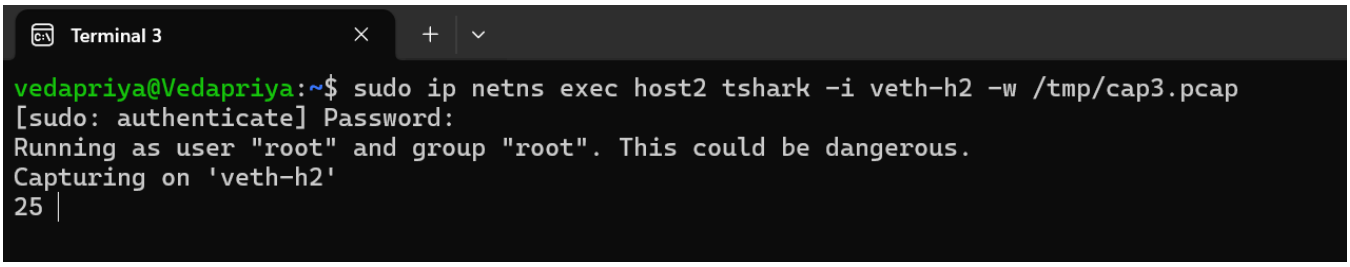
Terminal 1:



Terminal 2:



Terminal 3:



STAGE 12 — Fix permissions

sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap

STAGE 13 — Check TTL

tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

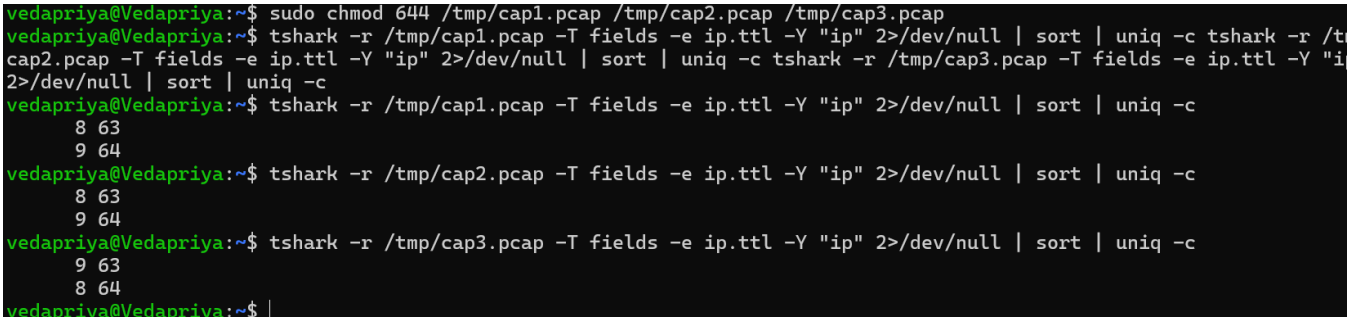
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

Expected:

cap1 → 64

cap2 → 64

cap3 → 63



STAGE 14 — Read your actual message from the capture

tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e data.text 2>/dev/null

You will see your actual message text extracted from the packet:

Hello from host1!

This is Veda

Computer Networks Lab

Three Point Capture Experiment

Message travelling through router

```
vedapriya@Vedapriya:~$ tshark -r /tmp/cap2.pcap -q -z follow,tcp,ascii,0

===================================================================
Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.0.1.1:39080
Node 1: 10.0.2.2:1234
18
Hello from host1!

13
This is Veda

22
Computer Networks Lab

31
Three Point Capture Experiment

33
Message traveling through router

        60
sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap

===================================================================
vedapriya@Vedapriya:~$
```

Connect from host1:

TERMINAL 1 — Setup (run all commands here)

Install tools:

sudo apt install -y tshark iproute2 netcat-openbsd wireshark

```
vedapriya@Vedapriya:~$ sudo apt install -y tshark iproute2 netcat-openbsd wireshark

[sudo: authenticate] Password:
tshark is already the newest version (4.6.4-1).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
iproute2 set to manually installed.
netcat-openbsd is already the newest version (1.234-1).
wireshark is already the newest version (4.6.4-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 206
```

When asked → choose **Yes**

Create 3 nodes:

sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2

Verify:

sudo ip netns list

```
vedapriya@Vedapriya:~$ sudo ip netns list
host2 (id: 2)
router (id: 1)
host1 (id: 0)
vedapriya@Vedapriya:~$
```

You must see: host1 router host2

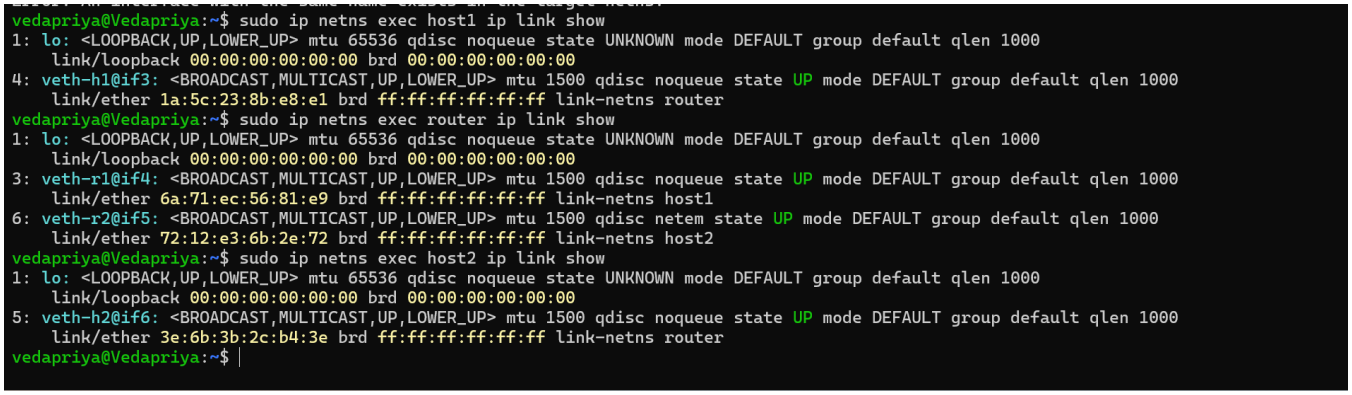
Create cables:

sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2

```
vedapriya@Vedapriya:~$ sudo ip link add veth-h1 type veth peer name veth-r1
vedapriya@Vedapriya:~$ sudo ip link add veth-r2 type veth peer name veth-h2
vedapriya@Vedapriya:~$
```

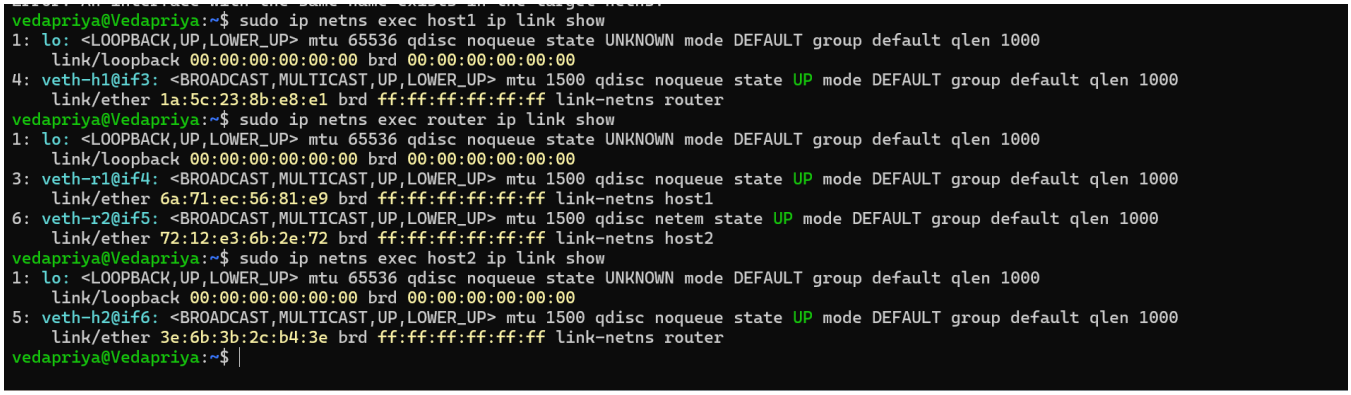
Plug cables into nodes:

sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2



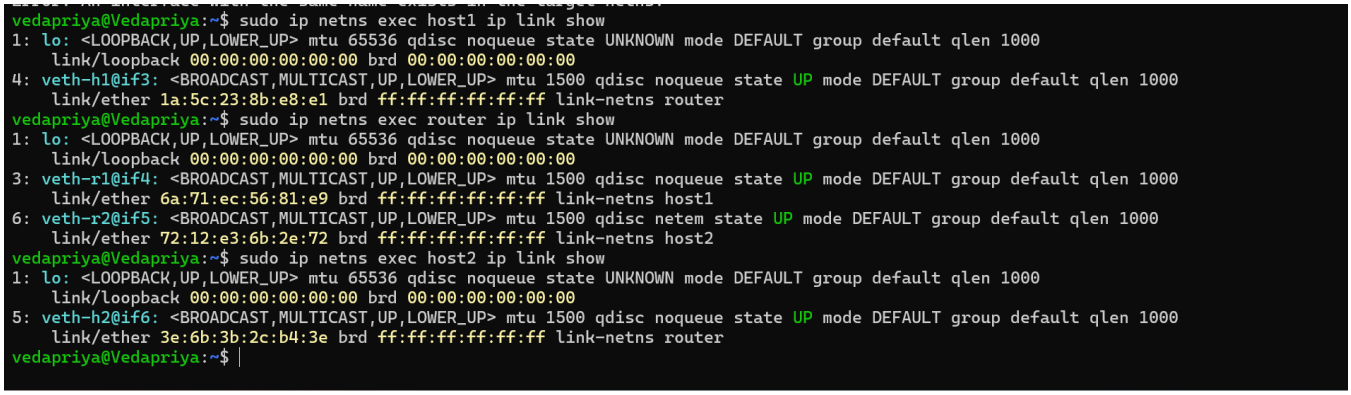
Configure host1:

sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up



Configure router:

sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up



Configure host2:

sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2

sudo ip netns exec host2 ip link set veth-h2 up

sudo ip netns exec host2 ip link set lo up

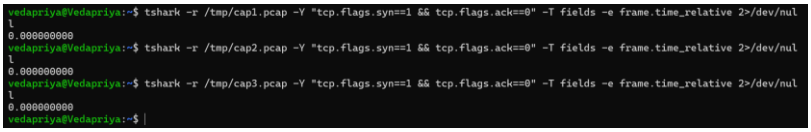
```
vedapriya@Vedapriya:~$ sudo ip netns exec host1 ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
4: veth-h1@if3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 1a:5c:23:8b:e8:e1 brd ff:ff:ff:ff:ff:ff link-netns router
vedapriya@Vedapriya:~$ sudo ip netns exec router ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
3: veth-r1@if4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 6a:71:ec:56:81:e9 brd ff:ff:ff:ff:ff:ff link-netns host1
6: veth-r2@if5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc netem state UP mode DEFAULT group default qlen 1000
    link/ether 72:12:e3:6b:2e:72 brd ff:ff:ff:ff:ff:ff link-netns host2
vedapriya@Vedapriya:~$ sudo ip netns exec host2 ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
5: veth-h2@if6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 3e:6b:3b:2c:b4:3e brd ff:ff:ff:ff:ff:ff link-netns router
vedapriya@Vedapriya:~$
```

Check delay:

tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null

tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null

tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null



Difference between cap1 and cap3 ≈ 20ms

Check retransmissions:

tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l

tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l

tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l

```
vedapriya@Vedapriya:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
vedapriya@Vedapriya:~$ tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
vedapriya@Vedapriya:~$ tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
vedapriya@Vedapriya:~$
```

cap3 shows highest count — loss happened between router and host2.

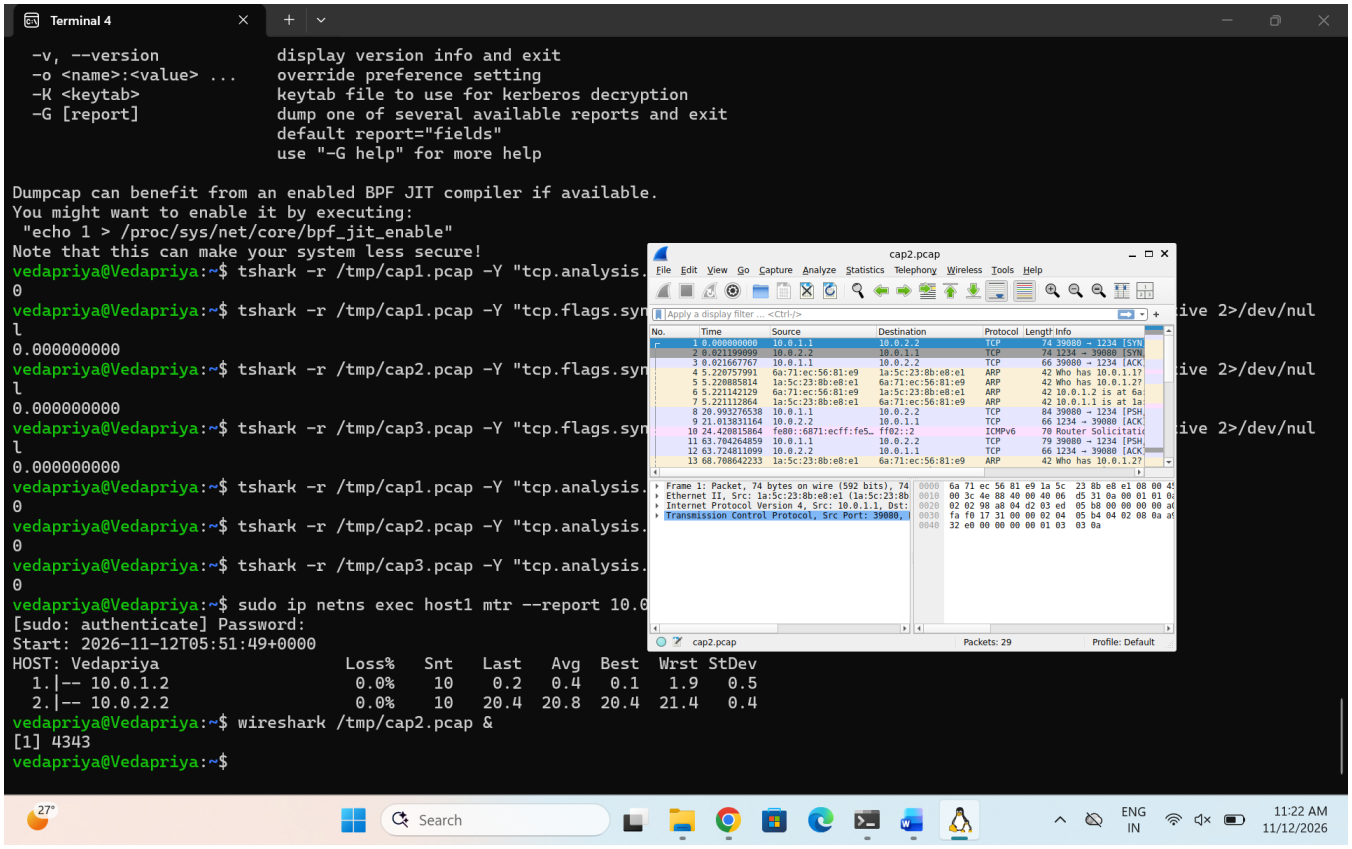
Check RTT:

sudo ip netns exec host1 mtr --report 10.0.2.2

```
vedapriya@Vedapriya:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
[sudo: authenticate] Password:
Start: 2026-11-12T05:51:49+0000
HOST: Vedapriya                   Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.2   0.4   0.1   1.9   0.5
  2.|-- 10.0.2.2                   0.0%    10   20.4  20.8  20.4  21.4   0.4
vedapriya@Vedapriya:~$
```

TERMINAL 4 — Open Wireshark

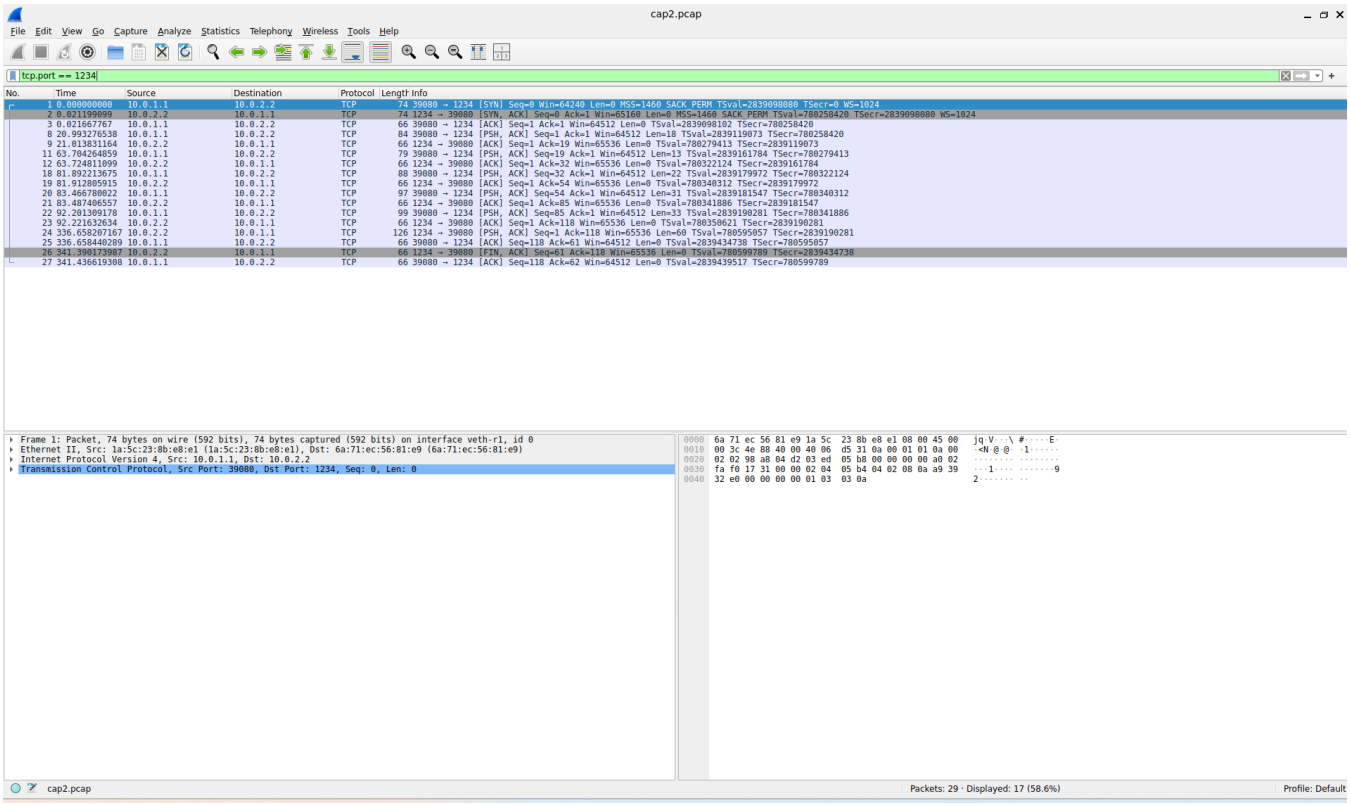
wireshark /tmp/cap2.pcap &



Inside Wireshark:

Filter your message only:

tcp.port == 1234

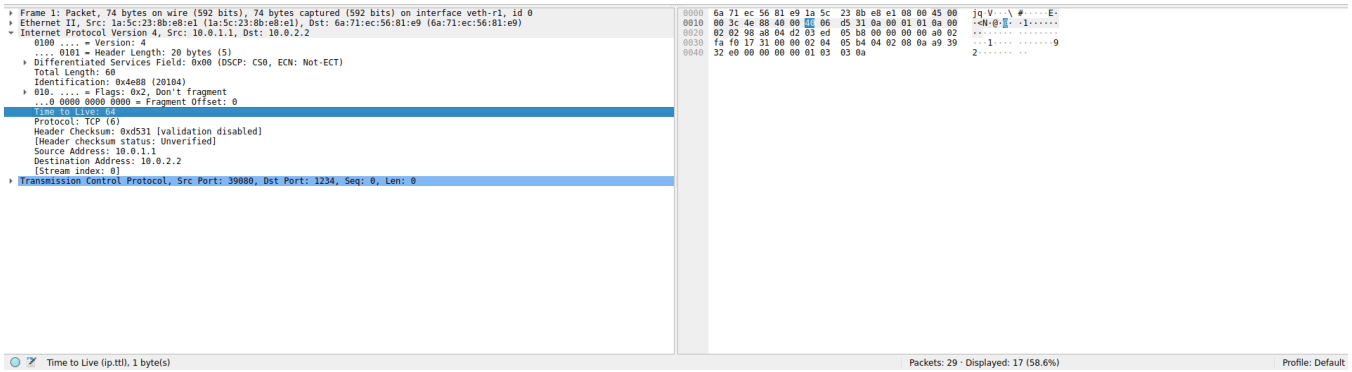


Click any packet → expand middle pane → you will see:

▼ Ethernet II

▼ Internet Protocol Version 4

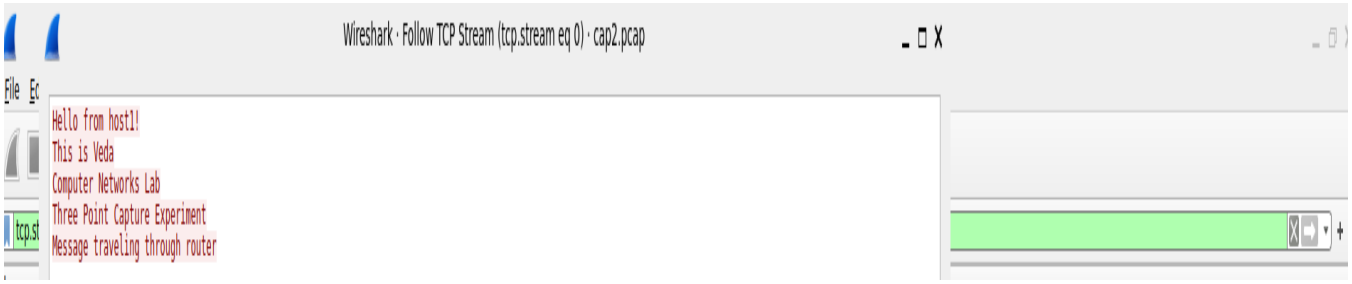
Time to Live: 64 ← TTL visible here



▼ Transmission Control Protocol

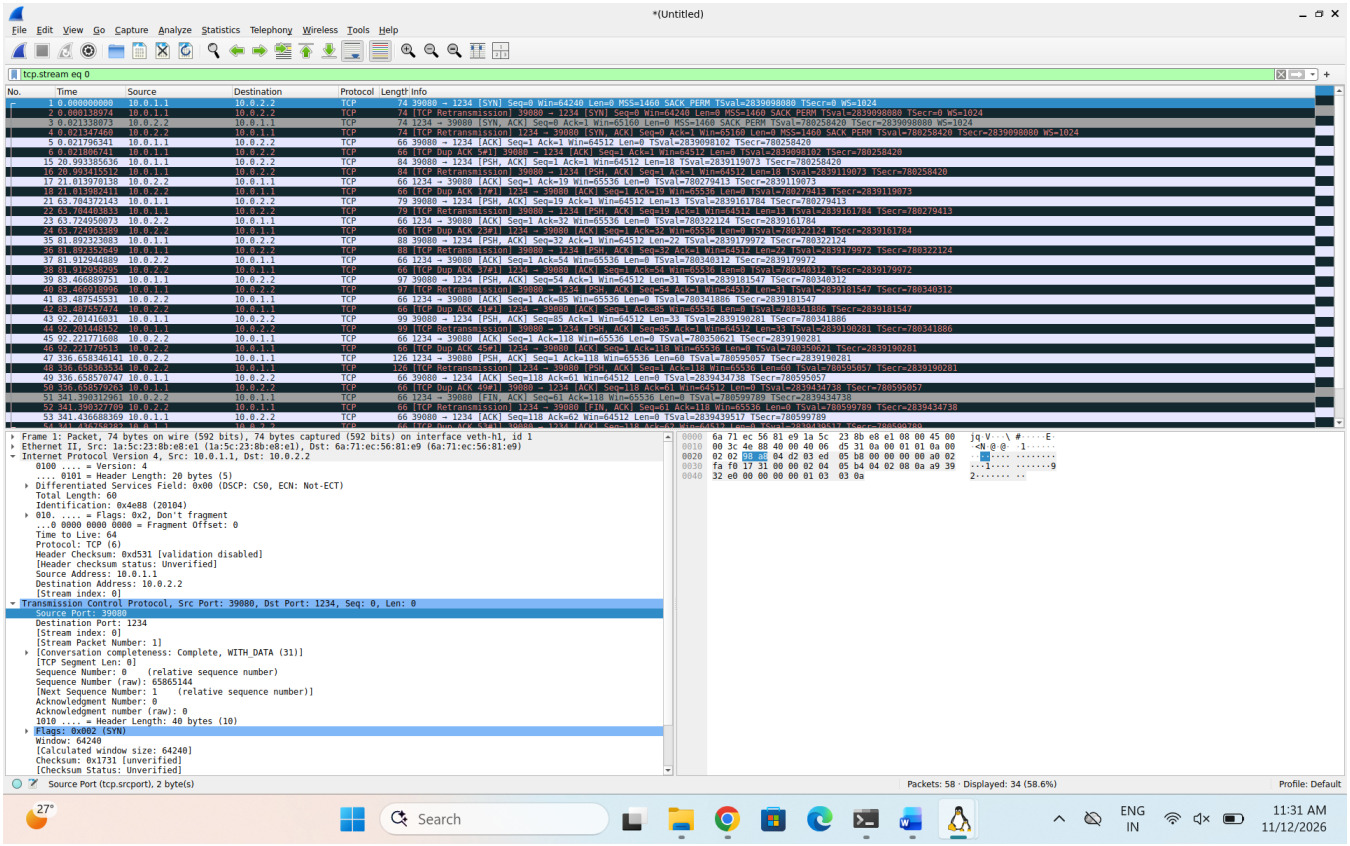
▼ Data

Hello from host1! ← your actual message visible here

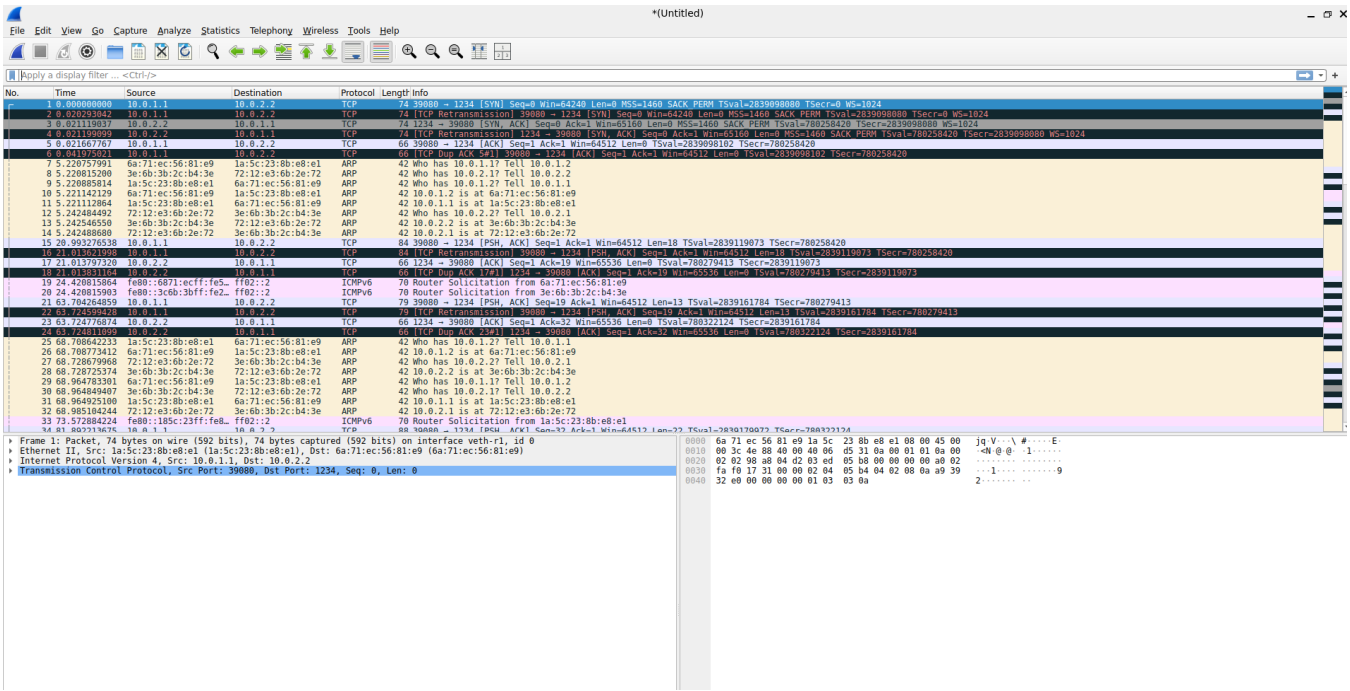


Merge all 3 captures:

File → Merge → select cap1.pcap



File → Merge → select cap3.pcap



Add TTL column:

Right-click column header

→ Column Preferences

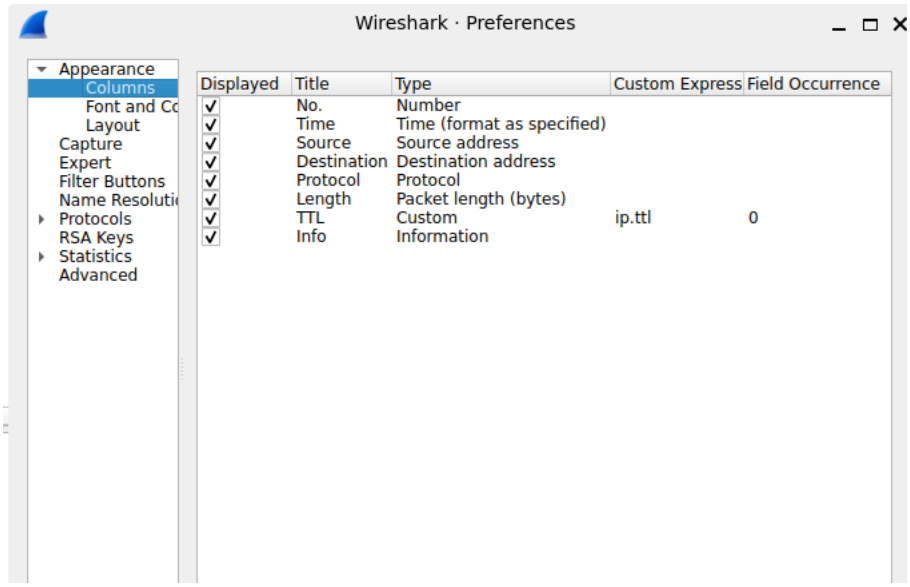
→ click +

→ Title: TTL

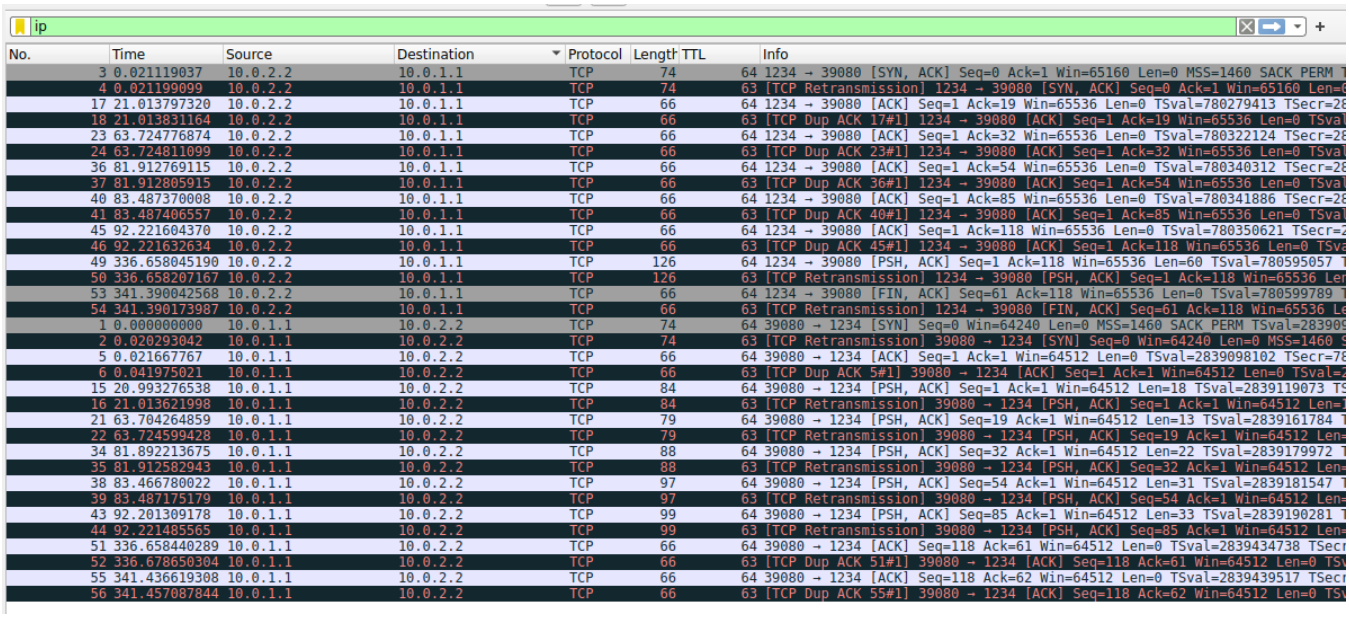
→ Type: Custom

→ Fields: ip.ttl

→ OK

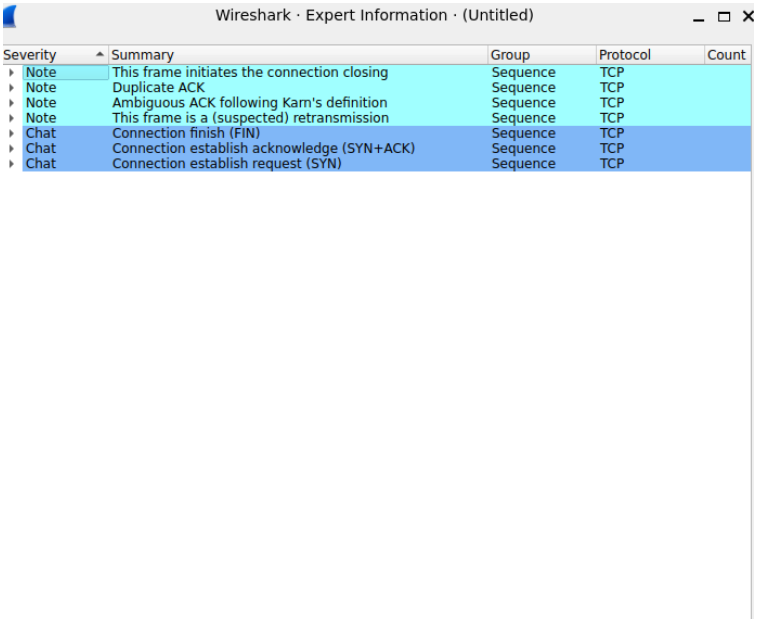


Students can now see TTL = 64 at points (1) and (2) and TTL = 63 at point (3)



Show packet loss:

Analyze → Expert Information



CLEAN UP — Terminal 4

sudo ip netns del host1

sudo ip netns del router

sudo ip netns del host2

```
vedapriya@Vedapriya:~$ sudo ip netns del host1
[sudo: authenticate] Password:
vedapriya@Vedapriya:~$ sudo ip netns del router
vedapriya@Vedapriya:~$ sudo ip netns del host2
```

Verify:

sudo ip netns list

Must return blank.

```
vedapriya@Vedapriya:~$ sudo ip netns list
vedapriya@Vedapriya:~$
```